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Knowledge graph construction for pipeline standards: Enhancing efficiency in the oil and gas industry

Quan Hui [a,b], Bing Liu [c], Jianchun Fan [d,*], Dehua Guo [e], Jiaru Chen [f], Xiao Tan [c], Zeheng Peng [c]

[a] College of Artificial Intelligence, China University of Petroleum-Beijing, Beijing, 102249, China
[b] Department of Science & Technology, China Oil & Gas Pipeline Network Corporation, Beijing, 100013, China
[c] Standardization Research Center, PipeChina Institute of Science and Technology, Tianjin, 300450, China
[d] College of Safety and Ocean Engineering, China University of Petroleum-Beijing, Beijing, 102249, China
[e] Sub-Institute of Public Security Standardization, China National Institute of Standardization, Beijing, 100191, China
[f] School of Information (National Characteristic Demonstration Software School), Xiamen University, Xiamen, 361001, China

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ABSTRACT

The intelligent development of oil and gas pipelines is crucial for overcoming the challenges of transitioning from informatization to intelligence, especially considering the vast number of standards governing pipeline operations. With over 3000 distinct standards collected, their multi-source heterogeneity makes effective manual use virtually impossible, leading to a significant semantic gap and difficulty in scenario-based retrieval. This study introduces a novel knowledge graph (KG) specifically designed to address this challenge by providing a unified, efficient method for managing and applying these standards. The KG is constructed based on a predefined ontology to provide an efficient semantic framework for technical specifications. We employ a novel hybrid information extraction framework, combining advanced neural models (BERT/BiLSTM-Attention) with rule-based methods and a Graph Convolutional Network (GCN) for relationship extraction. This approach resolves terminology ambiguity inherent in multi-source documents and accurately extracts entities and relationships from heterogeneous standards. This framework facilitates the construction of a comprehensive knowledge graph for pipeline network technical specifications, thereby enhancing the accessibility and usability of critical technical knowledge. The constructed KG, covering 17,655 entity nodes and 20,454 relationships, facilitates the intelligent retrieval of information and promotes the digital transformation of pipeline management. The validation demonstrates a high F1-Score in information extraction and confirms the capability of the KG to shift the knowledge retrieval paradigm from slow document searching to rapid, non-linear relational querying. By integrating data from diverse standards, this approach supports technological advancements, enhances safety and sustainability in the oil and gas industry.

1. Introduction

Pipeline transportation serves as the lifeline for the conveyance of oil and natural gas resources, playing a pivotal role in the entire oil production process. In light of the transformation from informatization to intelligentization, the development of pipeline intelligentization has emerged as an anticipated trend in the oil and gas pipeline industry. Characterized by high technology intensity, this trend aims to meet current demands. In 2017, PetroChina put forward the concept of "Intelligent Pipelines and Smart Pipeline Networks" and required, in terms of policy, to focus on strengthening the innovative integration of advanced technologies such as Internet of Things (IoT), big data, and cloud computing with oil and gas pipeline networks (Guo et al., 2022).

Unlike the situation in China, the concept of intelligent pipelines is not as prevalent in the global oil and gas pipeline sector, and comparable systematic frameworks have not yet been uniformly adopted. Nevertheless, the relevant technologies have already reached a high level (Wu and Zuo, 2020).

The traditional development and management model of oil and gas pipelines is confronted with multiple bottlenecks. On one hand, the management method relying on manual work leads to low efficiency, which not only pushes up labor costs but also increases operational

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* Corresponding author.
E-mail addresses: huiquan@pipechina.com.cn (Q. Hui), lbcat@126.com (B. Liu), fjc688@126.com (J. Fan), guodh@cnis.ac.cn (D. Guo), jruchen27@163.com (J. Chen), amoundtan@163.com (X. Tan), pengzh02@pipechina.com.cn (Z. Peng).

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risks due to the subjectivity and limitations of human decision-making. On the other hand, the complexity of the spatial distribution of physical pipeline networks, the functional correlation of supporting facilities, and the dynamic interference of the external environment further exacerbate the difficulty of management (Ai, 2013). Of particular importance is that pipeline safety is directly related to environmental safety, production safety, national energy security, the safety of people's lives and property, and even the overall stable development of the economy and society. Its significance determines the necessity and urgency of constructing a pipeline standard system (Guo et al., 2022).

Currently, there are significant differences in the regulatory systems and standard-setting models for oil and gas pipelines among countries worldwide. Taking China as an example, a four-level standard system has been formed, covering national standards, industrial standards, local standards, and enterprise standards. Some of these standards have referenced the specifications and requirements of international organizations such as the International Organization for Standardization (ISO) during their formulation process, ensuring the compatibility and scientificity of the standards.

However, pipeline standards face prominent industry pain points in practical application. Firstly, the distribution forms of standards are highly heterogeneous (including paper documents and digital formats such as PDF and XML), and standards from different sources (e.g., national ministries, industry associations, enterprises) exhibit obvious differences in the expression of technical requirements, resulting in great difficulty in the collaborative application of standards (Damagh and Faure, 2016). Secondly, the number of pipeline standards is enormous. Traditional knowledge management methods, such as keyword retrieval and document classification storage, are fundamentally limited as they are unable to systematically integrate these multi-source standards. More critically, they fail to accurately capture the deep contextual semantic associations between standard clauses. The semantic information hidden in standards (such as the constraint relationships between technical indicators and the complementarity of different standard clauses) cannot be effectively mined. This semantic gap forces technical personnel to spend a great deal of time manually screening information when retrieving standards for specific scenarios, leading to a high risk of missing key clauses, which directly results in low knowledge reuse efficiency and restricts rapid decision-making in pipeline engineering practice.

To address these limitations, this study proposes to construct a pipeline network technical requirement knowledge graph. As an efficient knowledge representation form, compared with traditional knowledge management methods, knowledge graphs can transform scattered standard clauses into a dynamic semantic network of "entity-relationship-attribute". This structure explicitly maps the relationships between technical indicators (e.g., the design requirements and applicable standard documents corresponding to a certain construction indicator), significantly improving the efficiency of knowledge access and reuse (Guan et al., 2019).

The core goal of the knowledge graph constructed in this study is to address the efficient management issue of more than 3000 multi-source heterogeneous standard documents. Through a hybrid construction method, we realize the structured integration of multi-source standards, ensuring that technical personnel can quickly retrieve technical indicators that accurately match scenarios in engineering practice (such as pipeline construction quality control and operational risk investigation), thereby reducing the error rate and time cost of manual document processing. Specifically, the technical advantages of this knowledge graph directly counter the aforementioned limitations:

Firstly, it has **multi-source heterogeneous adaptability**. Through format parsing and ontology mapping, it achieves the unified management of standard documents from different sources and in different formats.

Secondly, it possesses **semantic association mining capability**. Based on the relationship types defined by the ontology, it reveals the implicit constraint relationships between technical indicators (e.g., the matching relationship between "pipeline pressure level" and "pipe material selection"), directly addressing the challenge of unmined hidden semantics.

Thirdly, it features **scenario-based retrieval efficiency**. It supports associated retrieval based on engineering scenarios (e.g., "pipeline construction in alpine regions") and directly returns all technical requirements that need to be met in the scenario, rather than the isolated standard clauses returned by traditional keyword searches.

This study constructs this knowledge graph based on a predefined ontology, which defines entities, relationships, and attributes, providing an efficient framework for organizing and querying the technical specifications embedded in these standards (Section Methods). The proposed method adopts a top-down pattern layer construction, first establishing a knowledge framework for technical requirements in pipeline engineering, and then extracting information from different types of pipeline standards to form a comprehensive knowledge graph. This method can manage multi-source and heterogeneous pipeline standard documents, ensuring the accurate and rapid acquisition of relevant technical information, thereby realizing intelligent retrieval and decision-making in engineering practice.

To validate our approach, we first present quantitative experiments to evaluate the performance of our information extraction models (Section Experiments and Results). Furthermore, a case study (Section Case Study) demonstrates the practical advantages of the knowledge graph, showing that this method can quickly locate the technical indicators involved in specific application tasks and improve knowledge reuse and efficiency in the pipeline industry.

2. Related work

In this study, the construction of knowledge graphs for pipeline standards in pipeline intelligentization was explored, and related research works on Pipeline Intelligentization and Knowledge Graph are presented in this section.

2.1. Pipeline intelligentization

Pipelines are the arteries of the oil industry, and oil and gas pipeline intelligentization can optimize current complex systems, which is conducive to breaking many bottlenecks, such as high decision-making risks, high labor costs, and low management efficiency, that exist in the development of traditional oil and gas pipelines, and are conducive to the promotion of safe, efficient, and sustainable development. The concept of "intelligent pipeline and smart pipeline" was first proposed by PetroChina in 2017, and the China-Russia East-Route Natural Gas Pipeline Project was considered as a demonstration project for smart pipeline construction. Currently, smart pipelines have become a buzzword in China's oil and gas pipeline industry. Regardless of its definition, its core connotation is to satisfy various business needs through information management, decision-making support, and automatic operation, with the characteristics of high automation, intelligence, digitization, and networking at various stages of a pipeline's entire life cycle.

Internationally, with the advancement of Industry 4.0, Siemens has proposed the concept of "Pipelines 4.0" and developed a comprehensive toolset that supports pipeline asset construction, operation management, and full-life cycle optimization. As a core component of smart pipelines, its intelligent pumping software integrates advanced data analysis, artificial intelligence, machine learning and other technologies, which can comprehensively optimize pump station operation schemes, thereby improving load management, operational energy efficiency and batch planning levels (Wu and Zuo, 2020). In terms of intelligent leak detection, Cruz et al. proposed a gas pipeline leak detection and location technology based on machine learning and

acoustic methods, which can effectively identify tiny leaks in low-pressure pipelines and reduce false alarm rates (da Cruz et al., 2020). In addition, Li et al. developed a data-driven corrosion prediction model based on principal component analysis, artificial bee colony algorithm and support vector regression, which realizes high-precision simulation and prediction of corrosion rates in subsea pipelines (Li et al., 2021).

Although certain pipeline companies have certain high-level and new technologies in the construction of intelligent pipelines, the development of research activities related to intelligent pipeline networks is scattered without any industry leader, and intelligent pipeline network technology is still in the early stages of development (Liu et al., 2023). It is still of great significance for the exploration and application of different intelligent technologies in the pipeline industry. Intelligent pipelines require intelligent technologies to cover their entire life cycle, with key technologies in each stage. In the feasibility study stage, two types of methods are commonly used for the key parts of demand forecasting: qualitative and quantitative. Qualitative methods include the expert meeting method (Hanea and Nane, 2019) and the Delphi method (Duru et al., 2012), etc, whereas quantitative methods include the linear regression forecasting method (Lopes et al., 2020) and stationary time series prediction method (Shi et al., 2021), etc. In the design phase, most of the optimization processes for pipeline networks represent the various constraints and objectives in the design phase by constructing mixed-integer linear programming models (Hong et al., 2019), which can be solved using intelligent algorithms or piecewise linearization of nonlinear terms (Wang et al., 2018). The equipment optimization process typically uses support vector machines (Maral et al., 2019) and artificial neural network (ANN) models (Ekradi and Madadi, 2020) for compressor design. Furthermore, the mean flow rate method, singularity method utilizing the principle of the Fourier transform (Borges, 1990), and quasi-three-dimensional method (Peng et al., 2002) are used to design the pumps, and both the compressor and pump design are critical to the overall optimization of the pipeline. During the construction phase, various raw data and geological samples can be collected at observation points through IoT and 5G networks (Tsai et al., 2021) to achieve comprehensive monitoring of pipeline status from equipment procurement, construction, and completion to achieve better coordination of resources from all parties.

Since pipeline standards provide comprehensive guidelines on various aspects such as pipeline design, route selection, installation techniques, equipment quality, welding, fabrication, testing, and maintenance, these standards serve as foundational reference points throughout all stages where key technical methods are applied. The close integration of these methods with the standards ensures that each phase, from design to operation, is aligned with industry best practices. However, there has been limited research on how to effectively manage and reuse the knowledge embedded within these standards. This study, therefore, explores innovative methods and technologies for intelligent pipeline construction, using pipeline standards not only as a technical framework but also as a starting point for building a structured approach to knowledge management and reuse.

2.2. Knowledge graph

Knowledge Graph is an efficient knowledge representation method, which is a directed graph network comprising nodes and directed edges, where nodes in the graph represent entities or concepts and edges represent semantic relationships between different entities or between concepts. The basic constituent unit of a knowledge graph is a triple consisting of entity 1, a Relationship, and Entity 2, with entity 1 referring to the head entity, entity 2 referring to the tail entity, and relationships referring to the connection between the entities. Its core technology involves transforming complex data information into visualized graphs, which solves the problem of failing to rapidly and accurately obtain the required data because of excessive information. Compared with traditional knowledge management methods, it is conducive to inference and tracing between knowledge and can help users quickly and effectively obtain relevant knowledge, logical relationships, and information related to the knowledge.

The knowledge graph was first proposed in 2012 by Google Inc. to identify and organise entities in text, providing links to relevant entities, and thus improving the efficiency of search engines. Thereafter, many other companies, such as Microsoft, began to develop their own knowledge graphs. In recent years, knowledge graphs have become the foundation of many information systems, and significant progress has been made in the application of knowledge graphs to specific fields by exploiting their powerful ability to provide semantically structured information. In the field of medicine, Shi et al. successfully integrated health data into heterogeneous textual medical knowledge and provided an algorithm to prune meaningless inferences on a knowledge graph to improve inference performance (Shi et al., 2017). In the field of cyber-security, Qi et al. added an event ontology to define the relationships between attacks, events, and alerts when constructing a knowledge graph (Qi et al., 2018). Based on an association analysis algorithm, dynamic attacks can be effectively detected and predicted to protect cyber assets.

In the oil and gas industry, several researchers have begun to build knowledge graphs to mine potential information from data and highlight key insights, aiming to provide important references for decision-making across different issues. Knowledge graphs offer several key advantages, including enhanced data integration, improved decision support, and optimization of design and construction processes. However, to fully understand the potential of these technologies, it is important to reference real-world case studies that showcase their effectiveness.

One significant advantage of knowledge graphs is their ability to integrate data from various sources (e.g., sensor data, historical records, maintenance reports) into a unified structure, thus improving data management and analysis efficiency. For example, in a project focused on oil and gas pipelines, researchers employed knowledge graph technology to consolidate data from multiple sensors and used natural language processing (NLP) techniques to extract key information from maintenance reports (Cordeiro et al., 2024). This integrated knowledge graph enhanced data accessibility and usability, while significantly reducing the time needed for data analysis. Knowledge graphs also provide rich contextual information to support smarter decision-making. In complex engineering projects, they help engineers quickly locate relevant technical documents, historical cases, and expert recommendations. For instance, in an underground gas pipeline monitoring system project, researchers developed an intelligent software-hardware system based on knowledge graphs to monitor the pipeline system and cathodic protection equipment (Yuzevych et al., 2018). By integrating both historical and real-time data, the system offered more accurate failure predictions and maintenance suggestions, greatly improving the reliability and safety of the system. Additionally, knowledge graphs can assist in optimizing decision-making during the design and construction phases of complex projects. For example, in the Zhuhai Dahuangqinshan No. 1 Tunnel project, researchers used Building Information Modeling (BIM) and knowledge graph technologies to visualize and manage the construction process (Zhou et al., 2021). By creating 3D geological models and tunnel BIM models, the project team was able to more effectively plan both design and construction, reducing design changes and errors, and ultimately improving overall project efficiency.

These examples underscore the potential of knowledge graphs to transform the pipeline industry by enhancing data integration, supporting informed decision-making, and optimizing engineering processes. By applying these technologies in real-world projects, the pipeline industry can leverage the full benefits of knowledge graphs to improve efficiency and safety. Research on knowledge graphs can be divided into two categories: construction techniques and applications. Construction techniques focus on the extraction, representation, fusion, and inference of knowledge in the graph, and applications focus on applying

knowledge graphs to real systems and specific domains. Most of them apply knowledge graphs to question-answer systems, recommendation systems, and information retrieval systems (Zou, 2020). Thus, this study focuses on KG construction techniques of knowledge graphs. Based on the pipeline standard, after extracting entities and relationships from unstructured or semi-structured texts, they were correctly connected to the knowledge graph and new facts were inferred from them to enhance the value of knowledge.

3. Methods

In this section, the pipeline standards are first analyzed, focusing on the main technical contents and characteristics of typical application scenarios contained in the standard. Consequently, several elements and their interaction relationships related to the realization of specific business scenarios in the life cycle of oil and gas pipelines were summarized and extracted. The ontology definition was then completed, including the definitions of entity, property, and relation types. Information extraction was then performed on standards from different sources, and the extraction results were formed into a knowledge triplet (Entity 1, Relation, Entity 2) stored in the Neo4j database, realizing the construction of a knowledge graph of pipeline network technical specifications.

3.1. Ontology definition

Ontology refers to a structured way to represent entities and relationships in a domain and is the basic framework for describing and constructing domain knowledge (Hogan et al., 2020). A pipeline technical standard refers to a standard specifies with a minimum operational design level, and pre-construction, and post-construction of a pipeline and its components. It establishes the safety requirements for pipeline design, route selection, installation methods, equipment quality, welding, fabrication, installation, testing, operation, and maintenance. Standards are crucial reference points that guide the design, construction, and management of hazardous infrastructure such as high-pressure natural gas pipelines (Maslen and Ransan-Cooper, 2017).

Standards contain technical requirements that must be satisfied at particular phases of the construction of oil and gas pipelines, namely, technical indicators. These indicators are important for measuring, evaluating, and guiding project construction. The technical indicators contained in the standard are used to normalize projects based on specific business scenarios. Each standard object in a particular business scenario in pipeline construction, which can be a specific device or equipment, has certain aspects that constrain and normalize the specific objects to be measured and evaluated. These aspects are referred to as indicator objects. Throughout the entire regulation process, various factors affect the indicators to meet the required standards, including environmental, technical, and human factors. These are called indicator influencing factors.

A knowledge system for pipeline standards should be established based on the current professional status and industry practices in the field, utilizing existing mature professional classification results as much as possible to ensure practicality and usability (Liu et al., 2013/04). This study considered existing technical literature and industry reports, such as those from the ISO and American Petroleum Institute (API). In addition, practical application needs were considered, and a knowledge framework for the technical specifications of the pipeline network was constructed by recording and describing the technical indicators within the pipeline standards. These indicators were incorporated into two core components of the domain ontology: concepts and relations. The semantic understanding of the standards was enhanced to enable the framework to clarify the relationships between various technical indicators. Consequently, these relationships are more explicit and comprehensible. Fig. 1 illustrates the key elements of technical indicators and their functional relationships within the pipeline network standards.

To facilitate the presentation of data sources, the relevant data sources used in this study are listed in Table 1.

Based on the underlying mechanisms of the normative effect of the technical indicators in the pipeline standard, this study proceeds with the definition of ontology, encompassing the definition of entity types, property types, and relation types.

3.1.1. Entity type definition

(1) Standard entities
The standard entity represents the standard or specification in pipeline construction, which contains the operational design levels related to the pipelines and their components, and the requirements before and after construction. It is typically referenced by its standard number. These standards are essential to ensure the safe and stable operation of pipelines.

(2) Standardized object entities
Each standardized object entity represents a specific object in a specific business scenario of oil and gas pipeline construction, which can be a certain device, an equipment, a project, or an oil field. References are made through the object name, which concretizes the entities involved in the pipeline standards.

(3) Indicator object entities
An indicator object entity represents an aspect or attribute of a standard object. The indicator is named considering the particular aspect or attribute of the specific object being measured or evaluated and is used to clarify its representation. Indicators can be classified into different types according to their evaluation, such as quality, safety, and environmental indicators. Indicators have qualitative or quantitative characteristics. They have specific content for the measurement and evaluation of particular aspects, directly reflecting technical specifications. Furthermore, the realization of technical specifications can be affected by multiple factors such as environmental conditions, operator skills, and material quality. The identification and management of these factors are key to ensuring that technical specifications are achievable, thus helping to optimize the project design and construction process.

(4) Business object entities
A business object entity describes the business process or operation during the construction of oil and gas pipelines, including design, construction, inspection, and maintenance. Integrating technical indicators into specific business scenarios ensures that the established technical standards are satisfied at each business step, thereby improving the quality and efficiency of the entire project.

3.1.2. Property type definition

In the knowledge graph of the pipeline network technical specification, the emphasis is on the representation and identification of the technical indicators. Attribute value pairs (Liao et al., 2012) were established to unify the information. They are the key descriptions and supplements of the entities.

Four types of entities — standards, standardized objects, indicator objects, and business objects — were selected to define their respective properties. Among these, the properties of indicator objects are relatively complex. These properties include: (1) the indicator name, which defines the aspect being measured or evaluated; (2) the indicator properties, which describe the characteristics of the evaluation; (3) the indicator content description, which details the specific evaluation criteria; (4) the indicator influencing factors, which refer to the elements affecting the realization of the indicator; and (5) the indicator type.

For example, in the context of pipeline construction, an indicator object could be the Pipeline Welding Quality Index. This index represents the welding quality in pipeline construction and consists of several key properties. The indicator name, ‘‘Pipeline Welding Quality Index’’, identifies the aspect being measured—namely, the quality of

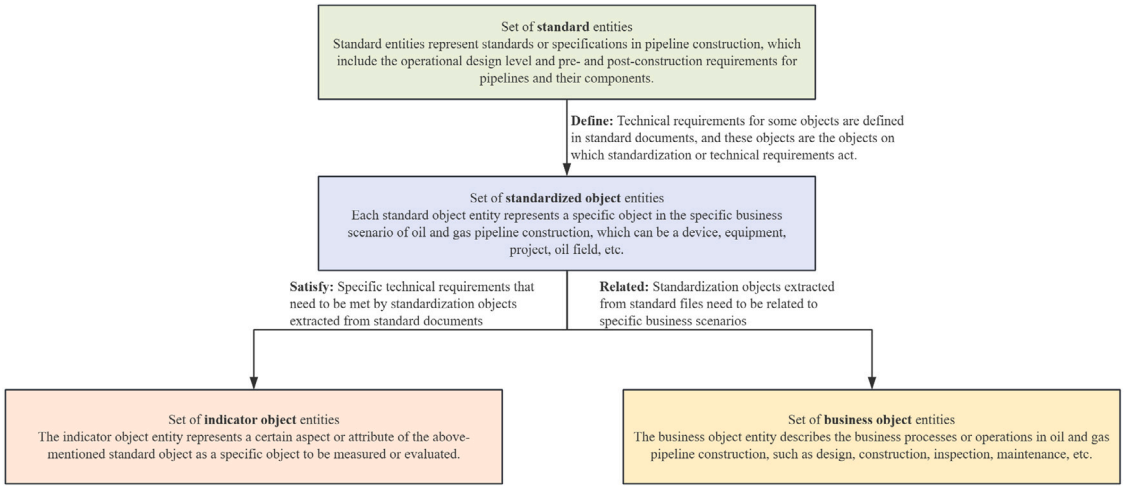


Fig. 1. Mechanism of the normative effect of technical indicators in the pipeline network standard.

Table 1
Relevant data sources used in this study.

Organization/Institute	Relevant Standard/Document	Description
ISO	Pipeline Construction Quality Management System	Specifies quality control requirements for pipeline construction
API	Oil and Gas Pipeline Specification	Provides design, construction, and operational guidelines for oil and gas pipelines
ISO	Steel Pipe Specification for Pipelines	Specifies technical and quality requirements for steel pipes used in oil and gas pipelines
API	Pipeline Welding Operations Specification	Provides standards for pipeline welding quality and procedures
ISO	Pipeline Operation and Maintenance Standards	Specifies technical requirements for pipeline systems during operation and maintenance
API	Pipeline Maintenance Standard	Provides standards for the maintenance of oil and gas pipelines

welding joints. Indicator properties might include characteristics such as seam strength, joint integrity, and the rate of welding defects. The indicator content description provides the specific evaluation criteria used to assess the welding quality, such as tensile strength tests, X-ray inspection results, and visual inspections of welding seams. The indicator influencing factors refer to the elements that can affect the achievement of the indicator, such as the welding technique used, the quality of materials, or environmental conditions during the welding process. Finally, the indicator type can be classified as either quantitative (e.g., measuring the defect rate as a percentage) or qualitative (e.g., determining compliance with standards based on expert assessment).

The definition of such indicators enables the standardization of critical aspects such as welding quality to ensure the consistency and accuracy of their application across various projects. This example demonstrates how indicator objects can contribute to the monitoring and management of technical quality, aligning them with broader standardization efforts.

Moreover, the properties of a standard include key details such as the International Classification for Standards (ICS) code, document code, date of issue, and date of implementation, which are essential for its use as a formal standard. Defining these properties is crucial to ensure that the standard remains accurate, traceable, and operable as a reference document in its application.

Based on the above summary, the common properties of the technical specifications of the pipeline network are obtained, as presented in Table 2.

The definitions of these properties allow detailed description and identification of the four types of entities in the pipeline network specifications. They provide key information to help understand the characteristics of each entity. In addition, such structured property representation also facilitates further extension of the description of existing resources, rendering the application and maintenance of knowledge graphs more flexible and efficient.

3.1.3. Relation type definition

After entity extraction, the pipeline network standard forms discrete entity nodes. To obtain specific semantic information, their relationship must be extracted and used to link the head and tail entities to form a net-like knowledge library. The relation types obtained in the technical specifications of the pipeline network are listed in Table 3.

(1) Defining relations
A defining relationship is required between the standard document and object entities. They indicate that standard documents define the technical requirements for certain objects, which are the objects on which the standardization or technical specifications are applied, that is, standard objects.

(2) Satisfying relations
A satisfying relationship exists between standardized object entities and indicator object entities. They indicate the specific technical requirements to be met by the standardized objects extracted from the standard document to ensure that the established technical standards can be satisfied in all steps of the project and to improve the quality and efficiency of the entire project.

(3) Related relations
A related relationship exists between standardized object entities and business object entities. They indicate that the standardized

Table 2
Entity-related attributes classification of entity related properties.

Entity name	Property name	Property description
Standard	standardName	Standard name
	ICS	International Classification for Standards (ICS) code
	CCS	China Classification for Standards (CCS) code
	documentCode	Document code
	documentType	Levels or categories of documents
	documentNumber	Document number
	IDTDocumentNumber	Identically adopted document number
	substituteRelation	Substituted relationship
	ChineseDocumentName	Chinese name or Chinese translation of the standardized document
	EnglishDocumentName	English translation of the standardized document
	adoptedDocumentNumber	Adopted document number
	adoptedEnglishDocumentName	Adopted English name for the document
	degreeOfConsistency	Identifier for the degree of consistency
	issueDate	Date of issue
	implementationDate	Date of implementation
	announcingBody	Issuing organization
Standardized Object	standardizedObject	Standard object name
Indicator Object	indicatorName	The name of the indicator
	indicatorType	Indicator type, i.e., the classification of the indicator
	indicatorQuality	Property of the indicator, qualitative or quantitative
	indicatorValue	Description of the indicator content
	indicatorInfluencingFactor	Indicator influencing factor
Business Object	businessObject	Name of the business object

Table 3
Types of relationships.

Head entity (e.g.)	Tail entity (e.g.)	Relationships
A standard document	Standard object (drilling operation)	Define
A standard document	Standard object (pipeline trunk)	Define
A standard object (drilling operation)	Indicator object (wellhead and casing protection)	Satisfy
A standard object (pipeline trunk)	Indicator object (pipeline crossing)	Satisfy
A standard object (drilling operation)	Business object (onshore oil and gas exploitation)	Related
A standard object (pipeline trunk)	Business object (oil and gas pipeline storage and transport)	Related

objects extracted from the standard document must be relevant to particular business scenarios, which helps understand the impact and scope of application of the standard for particular business practices.

3.2. Original data and processing

The two basic components of knowledge graph construction are the entity1, relation, entity2 triples and "entity-property(value)" key–value pairs. Entities are connected through their relationships to form a graph database (Wu et al., 2019). The most important step in knowledge graph construction technology is information extraction, and its completeness and accuracy directly affect the quality of the final knowledge graph. Information extraction is a technology that automatically extracts structured information, such as concepts, entities, events, and their properties and relationships, from structured, semi-structured, or unstructured data (Guo and He, 2015). This study developed the knowledge graph based on a previously defined ontology; that is, a set of entity types and relation types agreed upon in advance. Then, a rule-based method is used to extract information from the original text, while deep learning is used for complex parts.

3.2.1. Data preprocessing

First, the data were preprocessed. Standard documents require several preprocessing steps to facilitate subsequent extraction of entities and relationships. The standards collected in this study fall into three categories: national, industry, and enterprise standards. The formats of these documents include Word, PDF, XML, and JSON, and each format must be processed separately. The Apache POI library was used for reading and processing text-based standard documents. In addition, text cleaning and segmentation were performed to remove special characters and punctuation, address formatting issues such as line breaks and indentation, and divide the text into chapters, paragraphs, and other units. XML and JSON format documents are usually semi-structured and can be directly parsed using the respective parsers to read and analyze their content.

Due to the multi-source nature of the standards (national, industry, and enterprise), a significant challenge is the existence of terminology ambiguity and heterogeneity. For instance, the term "Maximum Allowable Operating Pressure" (MAOP) might be abbreviated differently or contextually imply related, but distinct, concepts like "Design Pressure". Our methodology addresses this critical issue through a two-pronged approach. Firstly, the use of BERT's intrinsic contextualization capability during the entity recognition phase is crucial. BERT's Transformer architecture allows it to generate deep, bidirectional, and context-aware vector representations (embeddings) for every word, making it highly effective at discerning the nuanced meaning of a term based on its surrounding technical language. This capacity is essential for automatically disambiguating polysemous terms and synonymous technical expressions within the specialized pipeline domain. Secondly,

the established domain ontology serves as a normative layer (Hogan et al., 2020). By strictly defining entities (e.g., Standardized Object, Indicator Object) and their authoritative relationships (e.g., Define, Satisfy), the ontology standardizes the extracted terms, forcing all semantic variants to map to a single, predefined concept in the knowledge graph. This combination ensures high semantic accuracy and resolves terminology inconsistency inherent in heterogeneous standards.

3.2.2. Entity recognition algorithm

(1) Entity recognition for word and PDF standard documents
A flowchart of the proposed entity recognition algorithm for Word and PDF standard documents is shown in Fig. 2(a) .
The specific steps are as follows:

(a) Obtain natural gas pipeline standard document data and preprocess it to generate a computer-readable text dataset.
(b) Use BERT and FastText to jointly encode the word vectors in the text and form a text embedding set. First, use BERT to encode the words in the text data into vectors. The BERT model consists of 12 layers of Transformer architecture, and its training process includes two stages: pretraining and fine-tuning. In the pretraining stage, BERT predicts missing or replaced words based on context, allowing it to effectively learn word vectors. In the fine-tuning stage, the pre-trained BERT model is adjusted for standard named entity recognition task, to optimize its performance. Then, FastText is used to encode the words in the text data into vectors as well. FastText generates word embeddings by learning the combination of word vectors and character n-grams. These word vectors are optimized to accurately reflect semantic information. Finally, the word vectors generated by BERT and FastText are concatenated to form a unified text-embedding set.
(c) Use a Bidirectional Long Short-Term Memory (BiLSTM) network and an attention mechanism to update the word vectors and learn the contextual dependencies between sentences. The BiLSTM network learns contextual information through both forward and backward propagation, enabling it to effectively capture long-distance dependencies. The attention mechanism enhances contextual relationship learning by calculating the importance of each word in the sentence and adjusting the word vector representation to better capture the key semantic parts of the sentence.
(d) Domain-Specific Justification for BiLSTM-Attention Architecture: In pipeline standards, technical requirements are often expressed in long, convoluted sentences containing multiple entities and constraints. Specifically, the BiLSTM-Attention architecture is crucial for this domain because:

- Handling Long-Range Dependencies: The BiLSTM layer effectively captures the non-local dependencies between entities that are syntactically distant within a single, complex standard clause (e.g., a **material** requirement stated early in a sentence applying to a **testing method** mentioned much later). This capability is superior to simple sliding-window approaches.
- Focusing on Key Indicators: The Attention mechanism is vital for assigning higher weight to core technical indicator phrases or constraint values (e.g., specific **pressure values** or **corrosion rates**), ensuring the model correctly prioritizes the most salient information for entity boundary determination and classification.

The combination of deep contextual embeddings (BERT) and this enhanced sequence modeling (BiLSTM-Attention) provides a robust solution for extracting entities from the complex and formal language of pipeline standards, a necessity validated by the significant performance improvement demonstrated in Section 5, where our model outperforms strong baseline architectures.

This approach enables the accurate identification of named entities in documents, with the ability to identify over 87% of entities in practical applications.

(2) Entity recognition for XML format standard documents
For XML format standard documents, entity recognition is achieved through XPath tag matching, ensuring efficient and precise content extraction. For explicitly defined tags, XPath is used to directly obtain the text within the tags as entity content. When dealing with standardized objects, pattern matching and parsing based on the markup conventions of the XML file are applied to ensure the extraction of the targeted information that adheres to specific rules.
(3) Entity recognition for JSON format standard documents
For JSON format data, large language models are used to identify entities such as “affiliated information”, ”organization names”, “preamble”, and ”standard objects”. The specific methods include the following:

(a) For easily identifiable entities like “affiliated information” and “organization names”, entity recognition is performed using In-Context Learning (ICL).
(b) For more complex entities like “standard objects”, local fine-tuning of a deployed large language model is employed for recognition.

This method achieved over 85% accuracy in entity recognition tasks and performed well.

3.2.3. Relation extraction algorithm

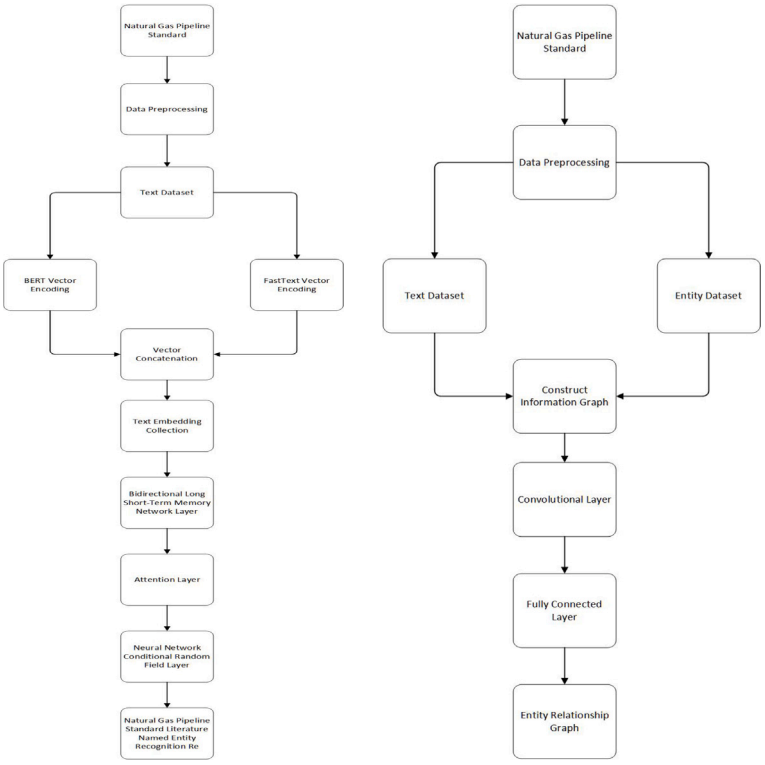
Various types of relations are extracted from standard documents. As shown in Fig. 2(b), a combined approach using rule-based methods and neural network models (such as graph convolutional networks (GCN)) is used to identify the types, names, and attributes of the relationships.

The specific steps are as follows:

1. Obtain natural gas pipeline standard document data, preprocess it, and generate a text dataset and an entity set.
2. Extract relations from the text dataset and entity set to form the corresponding entity-relation graph.

(a) Construct an information graph. Entities, sentences, and entity attributes are treated as nodes, with edges connecting entities to their attributes and entities to sentences, forming a heterogeneous information graph. The embedded representations of each node are learned using the BiLSTM model.
(b) Use GCN to update the embeddings of entity nodes. GCN aggregates information from neighboring nodes to enhance the representation of each node.
(c) Combine latent and structural representations, and use a fully connected layer for classification to predict whether a relationship exists between entities and determine the relationship type.

This model can identify over 75% of the entity relationships and effectively extract the relationships between nodes.



(a) Entity recognition algorithm flowchart. (b) Relation extraction model algorithm flowchart.

Fig. 2. Flowcharts of entity recognition and relation extraction algorithms.

3.3. Standards and information filtering strategy

To ensure the effective extraction of useful information and rejection of irrelevant information, the following filtering criteria were implemented:

1. Entity filtering criteria
   For entities in text data, we match predefined entity types such as "standard name". For complex entities like "standard objects", contextual learning methods are used for extraction, and fine-tuning is applied to improve accuracy.
2. Information filtering criteria
   For structured data (such as XML and JSON formats), matching is performed based on file tags and attribute rules to ensure that the extracted data conform to standard specifications.

3.4. Model evaluation and validation

To thoroughly evaluate the performance of the proposed extraction models, **Precision, Recall, and F1-Score** were utilized as the primary metrics, providing a comprehensive assessment of both the accuracy and completeness of the extraction results.

The definitions for these metrics are provided below:

- **Recall Rate (R):** Measures the proportion of actual positive instances correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **Precision (P):** Measures the proportion of positive identifications that were actually correct.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **F1-Score (F1):** Represents the harmonic mean of Precision and Recall, providing a single balanced measure of the model's overall performance.

$$\text{F1} = \frac{2 \times P \times R}{P + R}$$

Where TP (True Positive) represents the number of correctly identified entities or relationships, FN (False Negative) represents the number of actual entities/relationships missed, and FP (False Positive) represents the number of instances incorrectly identified.

Based on the comprehensive evaluation using these three metrics, the entity recognition model demonstrated robust performance across diverse document formats, achieving high **Recall** (e.g., over 87% for unstructured documents and 100% for structured formats like XML/JSON) to ensure knowledge coverage. Furthermore, it maintained high **Precision** and a balanced **F1-Score** across all test sets (detailed results are presented in Section 5.1.2). Similarly, in the relation extraction task, the GCN model successfully identified over 75% of the predefined semantic relationships with high F1-Score, demonstrating both coverage and quality in the extracted knowledge graph components. The consistent use of these three metrics ensures that the model evaluation is academically rigorous and fully accounts for both the coverage (Recall) and the correctness (Precision) of the generated knowledge graph elements.

3.5. Data storage

Finally, the samples were stored. In this study, the Neo4j graph database, an advanced graph database, was used. This database provides native graph data storage, retrieval, and processing and is characterized by good portability and scalability. Furthermore, it supports the storage of data volumes of tens of billions (Angles and Gutierrez, 2008). Here, 3524 pipeline standards were subjected to knowledge

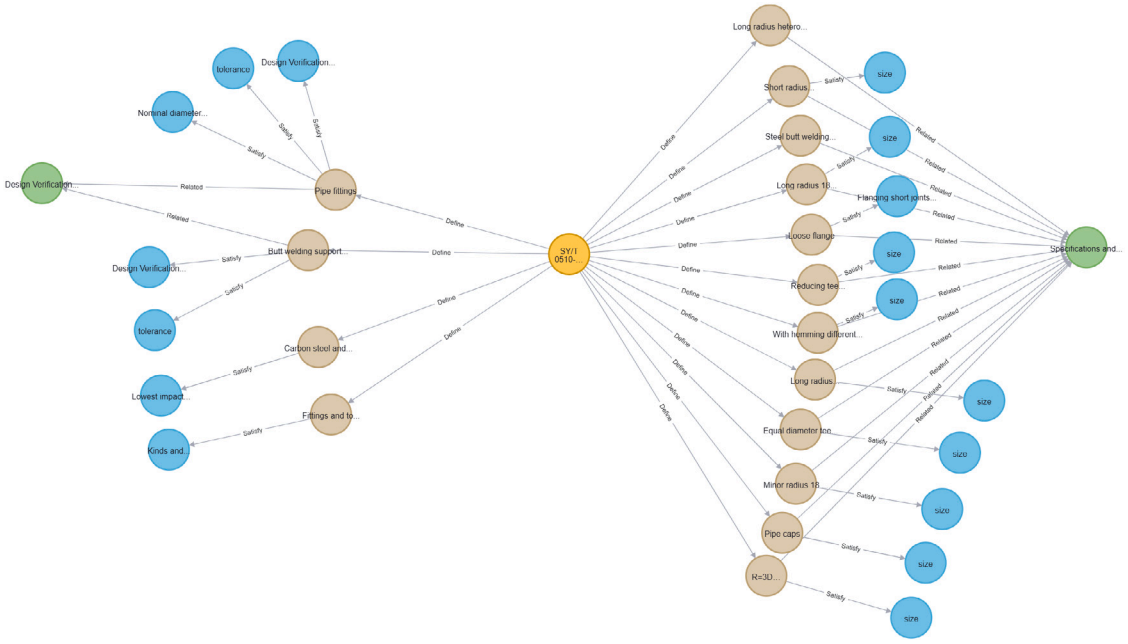


Fig. 3. Knowledge graph of pipeline network technical specifications (partial).

extraction to form 17,655 entity nodes and 20,454 interentity relationships. Entities and nodes were imported into the Neo4j graph database in the form of CSV files. The resulting knowledge graph of the technical specifications of the pipeline network is shown in Fig. 3. In the database, entity nodes or relation labels can be clicked to view the visually displayed knowledge graph data of pipeline network technical specifications, and specific technical specification information can be obtained by querying and traversing entities and relations in the database. Additionally, complex data analysis and mining can be performed using graph algorithms and query languages (e.g., Cypher) provided by Neo4j to discover potential patterns and associations. By exploiting these methods in graph databases, the information in the knowledge graph of pipeline network specifications can be deeply mined and utilized.

4. Case study

This section presents a case study to verify the feasibility and advantages of the pipeline technical specification knowledge graph. Currently, in pipeline engineering, compliance verification mainly relies on the personal experience and subjective judgments of technical personnel, which is time-consuming, inefficient, and prone to omissions or misjudgments owing to human factors. The pipeline technical specification knowledge graph proposed in this study addresses this issue by providing knowledge-driven decision support and intelligent search, thereby improving the reuse and efficiency of standard document knowledge.

4.1. Rapid understanding of technical requirements and improvement of compliance verification efficiency

In conventional pipeline engineering compliance verification, technical personnel must go through a large number of standard documents individually to check the technical requirements. However, this approach is inefficient and prone to errors. The pipeline technical specification knowledge graph can quickly locate all the standardized object nodes in a specific standard document, as shown in Fig. 4. These standardized object nodes represent objects in pipeline engineering that require standardization, such as equipment, materials, and processes, thereby significantly reducing the time and effort required for manual

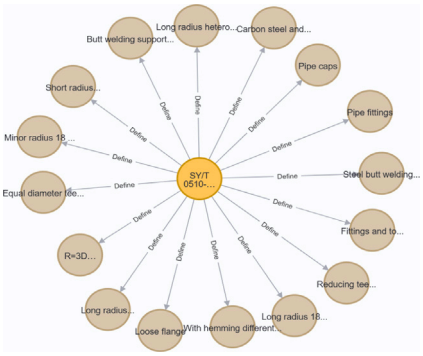


Fig. 4. Example of locating all standard objects included in the standard document.

inspections. Simultaneously, the completeness and accuracy of the knowledge graph ensure that all technical requirements are fully and accurately understood and applied.

After locating the standardized object nodes, the knowledge graph can further display all indicator object nodes related to that object. These indicator object nodes represent the specific technical requirements that the object must meet, such as pressure ratings, and material properties, as shown in Fig. 5. By reviewing these indicator object nodes, technical personnel can quickly understand all the technical requirements that the object needs to satisfy in the project, thus ensuring the project compliance.

4.2. Optimizing business management and decision-making

Understanding the technical requirements of various business objects is crucial for effective business management. However, quickly accessing information using conventional-knowledge management methods is challenging. Instead, the pipeline technical specification knowledge graph can quickly locate a business object node and view all the related standard object nodes, as shown in Fig. 6. This allows for a prompt understanding of all technical requirements related to the business, providing strong support for business management and

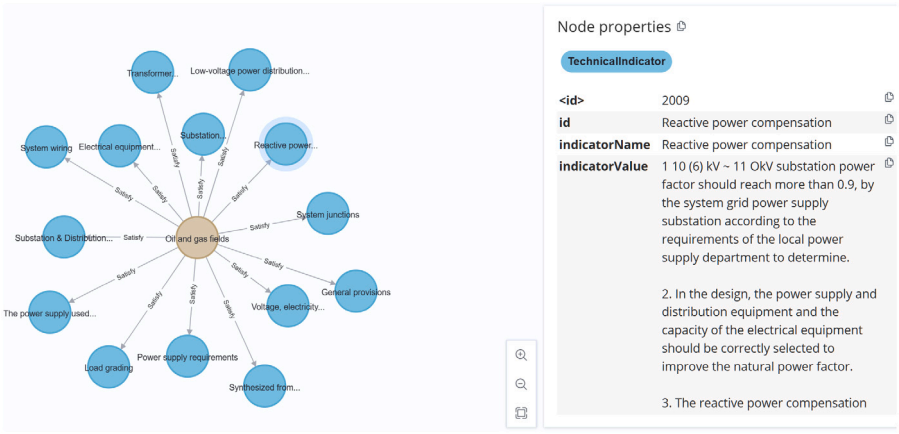


Fig. 5. Example of locating the indicator objects that the standardized object needs to meet.



Fig. 6. Example of locating all standardized objects related to the business object.

decision making. Simultaneously, all the associated nodes of a standard document can be viewed, as shown in Fig. 7, where the standard covers businesses such as anti-corrosion layer protection. Through this approach, all businesses covered by the standard can be comprehensively understood, thereby enabling better planning and coordination of various businesses.

5. Experiments and results

To systematically evaluate the proposed knowledge graph construction methodology, we designed a two-part experimental study. The first part focuses on the quantitative performance of our named entity recognition (NER) model against several established baselines. The second part provides a comparative evaluation of the practical efficiency gains achieved by using the constructed knowledge graph compared to traditional manual methods.

5.1. Entity recognition performance

5.1.1. Dataset and experimental setup

Dataset: Our dataset was derived from a corpus of pipeline standard documents, encompassing heterogeneous formats including Word, PDF, XML, and JSON. A subset of 200 representative documents, containing approximately 15,000 sentences, was manually annotated by three domain experts. This process yielded approximately 28,000 labeled entities across four predefined categories (Standard, Standardized Object, Indicator Object, and Business Object). The annotated dataset was randomly partitioned into training (80%), validation (10%), and testing (10%) sets.

Baseline Models: To provide a comprehensive performance benchmark, we selected three representative baseline models for comparison:

1. CRF: As a classic statistical model, Conditional Random Fields served as a baseline representing traditional machine learning approaches for NER.
2. BiLSTM-CRF: This model represents a standard and powerful deep learning architecture for NER, allowing for a direct comparison to evaluate the effectiveness of the attention mechanism in our model versus a conventional CRF decoding layer.
3. BERT-CRF: A strong baseline that leverages a pre-trained transformer model. Comparing against this model allows us to validate the contribution of the BiLSTM and attention layers integrated into our hybrid architecture.

Evaluation Metrics: We employed three standard metrics for the NER task: Precision, Recall, and F1-Score.

To ensure the robustness and reliability of our results, all experiments were conducted using a 5-fold cross-validation methodology. The reported scores are the mean values and standard deviations across the five folds. We performed a paired t-test to determine the statistical significance of the performance improvement of our model over the strongest baseline.

5.1.2. Results and analysis

The experimental results offer several key insights into the models' performance. A primary observation is the clear superiority of deep learning methodologies. All deep learning models significantly outperformed the traditional CRF model, which underscores their advanced capability to capture complex semantic features inherent in the technical language of pipeline standards.

Furthermore, the impact of pre-trained language models is evident. The BERT-based models (BERT-CRF and ours) achieved a substantial performance gain of approximately 5 percentage points in F1-Score over the BiLSTM-CRF model. This highlights the critical role of BERT's deep contextualized embeddings in accurately understanding and interpreting domain-specific terminology, which is a common challenge in specialized fields.

Finally, the results confirm the effectiveness of our hybrid architecture. Our proposed model achieved the highest performance across all metrics, reaching an F1-Score of 91.2%. The improvement of 1.8 percentage points over the strong BERT-CRF baseline was found to be statistically significant ($p < 0.05$). This validates our architectural design, where the BiLSTM layer effectively captures the long-range sequential dependencies common in standard documents, and the attention mechanism enables the model to focus on the most salient contextual information for a more precise entity classification. (see Table 4).

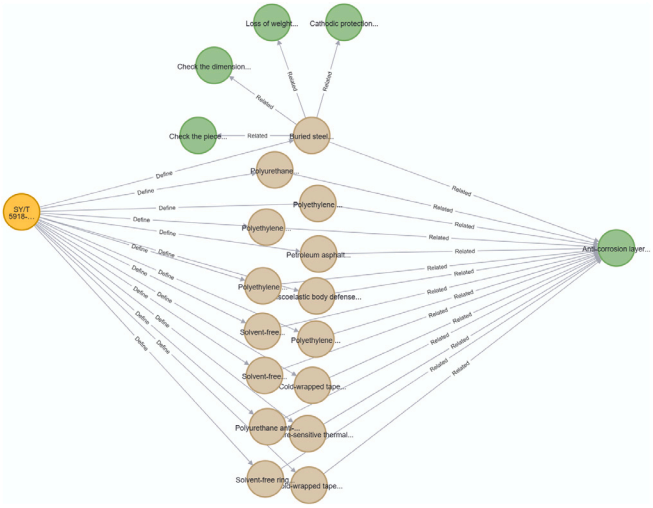


Fig. 7. Examples of locating all business objects involved in the standard document.

Table 4
Performance comparison of different models on the NER task.

Model	Precision (%)	Recall (%)	F1-score (%)
CRF	85.3 ± 1.2	81.7 ± 1.5	83.4 ± 1.3
BiLSTM-CRF	89.1 ± 0.9	86.5 ± 1.1	87.8 ± 1.0
BERT-CRF	91.4 ± 0.7	89.2 ± 0.8	90.3 ± 0.7
Our Model	**93.2 ± 0.6**	**91.5 ± 0.7**	**92.3 ± 0.6**

Table 5
Performance comparison of different models on the RE task.

Model	Precision (%)	Recall (%)	F1-score (%)
PCNN	79.5 ± 2.2	77.1 ± 2.4	78.3 ± 2.3
R-BERT	86.4 ± 1.6	84.0 ± 1.8	85.2 ± 1.7
Our Model	**88.2 ± 1.4**	**86.9 ± 1.5**	**87.5 ± 1.3**

5.2. Relation extraction performance

5.2.1. Dataset and experimental setup

Dataset: The dataset for relation extraction was constructed based on the entities identified in Section 5.1. For each sentence containing at least two entities, the entity pairs were manually annotated with one of the three predefined relation types (Define, Satisfy, Related) or a No Relation label. The final dataset comprises approximately 11,500 relation instances, which was partitioned using the same 8:1:1 ratio for training, validation, and testing.

Baseline Models: We selected three baseline models for relation extraction:

1. PCNN (Piecewise Convolutional Neural Networks): A classic and effective deep learning model for relation extraction that utilizes piecewise max-pooling to capture the structural information between two entities in a sentence.
2. R-BERT (Relational BERT): An adaptation of BERT for relation classification, which fine-tunes the pre-trained model by adding special markers around the target entities to focus on their relationship.

Our Model: As described in the methodology, our approach utilizes a Graph Convolutional Network (GCN) over the sentence's dependency parse tree to explicitly model syntactic constraints and long-range dependencies between entities.

Evaluation Metrics: The performance was evaluated using macro-averaged Precision, Recall, and F1-Score across the three primary relation types.

5.2.2. Results and analysis

The results presented in the table lead to several important conclusions. First, both BERT-based models (R-BERT and our GCN model) significantly outperform the PCNN model. This indicates that the deep contextual representations provided by BERT are crucial for accurately discerning the subtle semantic distinctions required for relation classification in complex technical texts.

Second, our GCN-based model achieved the highest F1-Score of 87.5%, surpassing the strong R-BERT baseline by 2.3 percentage points. This improvement underscores the value of integrating syntactic information into the modeling process. While R-BERT relies solely on the sequential text and entity markers, our GCN model leverages the sentence's dependency parse tree. This allows the model to more effectively capture long-range and non-sequential relationships between entities, which are common in the formal and often convoluted sentence structures found in standard documents. By propagating information along the syntax graph, our model can better filter out irrelevant contextual words and focus on the direct syntactic path connecting the entities, leading to a more robust and accurate relation classification. (see Table 5).

5.3. Discussion on practical efficiency and applicability

While the quantitative results demonstrate the high accuracy of our knowledge extraction, the practical value of the constructed knowledge graph is primarily realized through enhanced operational efficiency, which we demonstrate through a theoretical analysis of the structural advantages.

Traditional methods for pipeline compliance verification rely on manual retrieval and cross-referencing across vast, heterogeneous document repositories (e.g., thousands of pages of standards). This process is inherently time-consuming and error-prone. Our knowledge graph fundamentally addresses this inefficiency through three mechanisms:

1. **Integrated Semantic Network:** By converting fragmented, unstructured documents into a unified Neo4j graph structure, the system eliminates the need for multi-document keyword searching. All knowledge points are interconnected by semantic relationships.
2. **Direct Associative Retrieval:** The graph structure supports highly efficient, scenario-based querying (e.g., using a dedicated graph query language). A single query can traverse multiple paths to instantly locate all technical requirements associated with a specific business object (e.g., pipeline type, location, or operating pressure), a task that requires numerous sequential searches and human synthesis in the manual process.

3. **Consistency and Completeness Check:** The formalized, structured representation of the graph intrinsically aids in identifying missing or conflicting information, which dramatically reduces the time spent on validation and reconciliation of requirements—a major bottleneck in manual compliance checks.

In summary, the knowledge graph shifts the knowledge retrieval paradigm from slow, linear document searching to rapid, non-linear relational querying. This change in structural complexity provides a compelling theoretical basis for a significant increase in knowledge reuse efficiency and accelerates compliance-based decision-making in the oil and gas sector.

6. Conclusions

This study explored the application of intelligent technology in pipeline systems in the context of the transformation from informatization to intelligence. By introducing the concept of a knowledge graph, potential issues in understanding and applying pipeline standard documents can be addressed, specifically overcoming the semantic gap and the difficulty of scenario-based retrieval inherent in multi-source heterogeneous standards. Based on 3524 multi-source heterogeneous oil and gas pipeline standard documents, a unified knowledge representation method is used to define the entity, property, and relation types, with a defined ontology encompassing Standard, Standardized Object, Indicator Object, and Business Object entities. Information extraction and persistent storage of the results were achieved based on predefined ontologies, and a pipeline technical specification knowledge graph was successfully constructed. The extraction process employed a multi-strategy approach: entities were accurately identified using a BERT-FastText/BiLSTM-Attention network, and relationships were then captured by a Graph Convolutional Network (GCN) model applied to a constructed heterogeneous information graph, achieving an F1-Score of 87.5% on the relation extraction task. This knowledge graph covers 17,655 entity nodes and 20,454 relationships between the entities. Moreover, the Neo4j graph database was used to store the information visually, significantly improving the comprehensive and accurate understanding of technical requirements in standard documents. This study verified the feasibility and advantages of knowledge graphs in pipeline engineering management, demonstrating that this approach shifts the knowledge retrieval paradigm from slow, linear document searching to rapid, non-linear relational querying. This transformation significantly enhances the reuse and efficiency of standard knowledge and strengthens the efficiency and quality of engineering project management, particularly by accelerating compliance-based decision-making. However, the definition of entities and relationship types, as well as the completeness of the knowledge graph, still rely heavily on the quality of manual annotation and the current extraction framework's coverage. To further realize the full potential of this approach, future work will focus on developing more automated and zero-shot learning methods for relation extraction to improve generalization across new or unseen standards, and integrating real-time operational data to support dynamic decision-making during the pipeline's operational life cycle. This study provides strong support for the intelligent transformation of the oil and gas pipeline industry and points out directions for further research and applications. With continuous technological advancements and expanding applications, we consider that knowledge graphs will contribute significantly to the future oil and gas pipeline industry.

CRediT authorship contribution statement

Quan Hui: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Data curation. **Bing Liu:** Writing – review & editing, Validation, Supervision, Resources, Project administration. **Jianchun Fan:** Writing – review & editing, Investigation, Formal analysis. **Dehua Guo:** Writing – review & editing, Project administration, Investigation, Conceptualization. **Jiaru Chen:** Writing – review & editing, Project administration, Methodology, Investigation. **Xiao Tan:** Writing – review & editing, Validation, Supervision, Data curation. **Zeheng Peng:** Writing – review & editing, Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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